

Draft MSc dissertation: Uncertainty-Aware Calibration of Chlorine Wall Decay in the Net3 Benchmark Water Distribution Network

Note the length constraint of the MSc dissertation – I believe around 30 pages equivalent from the word count you are given? Your current draft is too long.

Generally, a coherent workflow; however, the key objectives are not fully supported by the results/evidence and discussion.

1. Overall assessment

The draft describes a workflow: forward WNTR/EPANET chlorine simulation, a synthetic three-zone wall-decay “truth”, noisy synthetic observations, deterministic grid-search recovery, GLUE behavioural calibration, ensemble propagation, and a network-wide threshold-risk map.

The writing is clear, the structure is logical, the limitations section is OK, and the choice to end at an operationally interpretable risk outcome rather than a calibrated coefficient is the right direction.

The sensitivity test is a genuinely good idea and produces the draft's most defensible result.

The central problem is that the headline calibration claim is not supported by the numbers reported.

For two of the three wall-decay coefficients the GLUE “behavioural distribution” is, to within rounding, the prior; the agreement between the weighted means and the imposed true values is an artefact of the prior ranges having been centred on those true values. The draft in places reads this coincidence as successful recovery. Section 3 sets out the arithmetic. This is fixable: largely by re-wording claims and running two additional low-cost experiments, but it must be fixed, because as written the strongest sentences in the Results are the least defensible ones.

The second problem is length: the draft is roughly double the permitted extent. The third is that the draft has not yet absorbed several method components which I suggest you include (see the enclosed Jupyter Notebook): systematic sensor error, censored likelihoods, error autocorrelation, and a priori identifiability analysis - all of which bear directly on its own conclusions.

2. What is working and should be preserved

- The workflow logic is sound end-to-end, and each stage genuinely passes uncertainty to the next rather than collapsing to a point estimate. This is the project's key contribution and should be stated more confidently.
- The identifiability gradient that grouped coefficients are not equally recoverable, and that recoverability depends on hydraulic visibility to the sensors rather than on the magnitude of the coefficient, is the correct conclusion. It is consistent with the foundation notebook's finding that parameter information is highly concentrated and peaks at intermediate residence time.
- The interleaved pipe-group experiment is well conceived, and the accompanying caveat that it changes both spatial arrangement and group sizes (39/39/39 versus 14/37/66) is exactly the right approach.
- The high-risk nodes identified (131, 243, 141, 139, 143) coincide with the longest-residence junctions found independently in the foundation notebook's zero-order tracer analysis (nodes 243 and 131 at

36.0 h and 34.6 h). This cross-validation is worth one sentence in the Results: it is free corroboration from an independent method.

- The limitations section is OK, correctly names the inverse issue, and cites Kaipio and Somersalo. Keep this.
- Figure quality is generally good.

3. Priority 1 — must be corrected before submission

3.1 The GLUE ensemble is, for two of three parameters, indistinguishable from the prior

Using the sampling ranges stated in the draft and the weighted statistics reported in §4.4:

Group	Prior range (m/day)	Prior sd	Behavioural sd (reported)	Width retained	Prior midpoint	Behavioural mean	True value
old	[−1.5, −0.2]	0.375	0.32	85%	−0.850	−0.960	−1.00
average	[−0.2, −0.04]	0.046	0.045	97%	−0.120	−0.120	−0.10
new	[−0.10, −0.005]	0.027	0.027	98%	−0.0525	−0.053	−0.05

Read the last three columns together. For the average group the behavioural mean equals the prior midpoint to three decimal places, and for the new group to within 0.02 prior standard deviations - while the imposed true values sat 0.43 and 0.09 prior standard deviations away from those midpoints respectively. The data moved these parameters not at all. Their apparent agreement with the truth is a property of the prior ranges chosen, not a result of the calibration.

The one group whose prior was not centred on its true value: old, midpoint −0.85 against truth −1.00, is the only one whose behavioural mean moved (by 0.29 prior sd, toward the truth) and the only one whose distribution narrowed appreciably (to 85% of prior width). That contrast is itself the cleanest identifiability result in the draft and should be presented as such.

Required changes. (i) Delete or heavily qualify the sentence “The weighted mean of every group lies close to its true value” and any construction that treats mean agreement as evidence of recovery. (ii) Add the table above, or its equivalent, reporting prior-to-behavioural width ratios for all three groups - this is a summary statistic for a GLUE analysis. (iii) State plainly that the average and new coefficients are not informed by the observations, rather than describing the average group as “only weakly constrained”. The draft already says this of the new group; the arithmetic says it of two.

Recommended experiment. Re-run the GLUE analysis with all three priors deliberately displaced so that no prior midpoint coincides with its true value: for example shifting each range by one prior standard deviation. Report how far each behavioural mean travels back toward the truth. This demonstrates a controlled identifiability experiment and would strengthen the dissertation.

3.2 The behavioural threshold cannot discriminate

The threshold is $RMSE < 0.12$ mg/L against observation noise of $\sigma = 0.10$ mg/L, and 1,685 of 2,000 samples (84%) are retained. That retention rate is a property of the threshold, not of the parameter space. With $N = 6$ monitors $\times$ 49 post-warm-up hours = 294 residuals, the sampling standard deviation of the RMSE statistic evaluated at the true parameters is

$$sd(RMSE) \approx \sigma / \sqrt{2N} = 0.10 / \sqrt{588} = 0.0041 \text{ mg/L},$$

so a threshold at 0.12 mg/L sits about 4.9 standard deviations above the noise floor. A parameter set at the truth exceeds it with probability of order 10^{-6} . The filter therefore rejects only parameter sets that are grossly wrong, which is why the retained ensemble is essentially the prior.

Required changes. Justify the threshold against the sampling distribution of the objective rather than against σ alone: a 95% acceptance band would be approximately $\text{RMSE} < 0.107 \text{ mg/L}$, and report retention rates for at least two thresholds so the reviewer can see the sensitivity. If a tighter threshold retains too few samples to propagate, that is itself a reportable finding about the information content of the monitoring array.

3.3 The experiment cannot detect structural error, and the search was centred on the answer

The truth is generated with three pipe groups and fitted with the same three pipe groups, using the same solver, timesteps and hydraulics. This is a complete inverse crime, not a partial one: there is no structural discrepancy anywhere in the experiment. The grid-search ranges compound this — with a $7 \times 7 \times 7$ grid over ranges chosen around the true values, the nearest grid nodes to the truth lie 0.007–0.067 m/day away, so “recovery to the nearest grid node” is close to guaranteed by construction. The reported ten-realisation spread for the old group ($\pm 0.12 \text{ m/day}$) is roughly half a grid step, i.e. largely quantisation.

This matters because the foundation study's central result is that structural error (fitting a parameterisation simpler than reality) produces a precise but biased estimate whose offset no amount of data removes. The draft's design excludes that phenomenon entirely, yet the Discussion generalises about calibration reliability.

Recommended experiment (I think the single most valuable addition). Generate one synthetic truth with finer heterogeneity than the fitted model: for example per-pipe coefficients drawn around the three group values, or a five-group truth, and calibrate the three-zone model against it. Report how far the behavioural ensemble sits from any simple average of the true field. This gives the dissertation a genuine structural-error result and directly addresses the inverse-crime limitation it already acknowledges.

3.4 Unit inconsistency between text and figures

The text correctly states that first-order EPANET wall coefficients carry units of length per time and reports them in m/day. All three GLUE distribution figures label their axes “kw_old (/day)”, “kw_avg (/day)” and “kw_new (/day)”. The lumped noise-sensitivity coefficient is also reported in day^{-1} while being described as a wall coefficient. Regenerate the figure axis labels and audit every reported coefficient for units. Note that this error originates in the foundation notebook, where the one-dimensional estimation section labels the wall-coefficient grid “1/day”; it is being corrected there as well.

3.5 The noise-sensitivity check is imported from a different experiment

The values reported in §3.5.2 and §4.3 (means of -1.147 , -1.138 , -1.143 day^{-1} with standard deviations 0.030, 0.052, 0.087 at $\sigma = 0.02, 0.05, 0.10 \text{ mg/L}$) are taken from the foundation notebook's global single-parameter experiment. That experiment uses a different monitor set, a different pipe grouping (five length-based groups), a different synthetic truth and a lumped coefficient. The draft acknowledges the lumped/grouped distinction but still describes the noise realisations as being added to “the same underlying true chlorine time series”, which is not the case here.

Required changes. Either re-run the noise-sensitivity sweep inside the three-zone setup, reporting how the behavioural width of each grouped coefficient scales with σ — which would be far more relevant — or

state explicitly that the check is inherited from a separate configuration and name the differences. The first option is preferable and is not expensive.

4. Priority 2 - alignment with the shared Jupyter Notebook

The following components exist in the project's foundation and operational notebooks and bear directly on the draft's own conclusions. Not all will fit within 30 pages; the ordering is by value per page.

#	Component	Why it matters here	Suggested extent
1	Systematic sensor error (bias, drift)	At the ± 0.1 mg/L class the systematic term dominates the error budget. A $+0.05$ mg/L offset at one informative monitor shifts the estimate by more than the entire random spread, and the effect is concave, halving the offset retains $\sim 70\%$ of the bias. The draft models random noise only and lists bias as an unquantified limitation.	~ 0.75 page + 1 small table
2	A priori identifiability (Fisher information, CRLB, profile likelihood)	Converts the draft's central finding from an observation into a prediction: compute which coefficients are identifiable before calibrating, then confirm with GLUE. The operational analysis does exactly this and obtains eigenvalues spanning three orders of magnitude for three zonal coefficients.	~ 1 page + 1 table
3	Censored likelihood	28 of 438 observations were clipped at zero, and the clipping concentrates in the low-residual old zone, precisely where the risk map is decided. The draft clips the data but then uses unweighted RMSE, which is misspecified there.	~ 0.5 page
4	Error autocorrelation	All intervals assume hourly independence. Real records are strongly autocorrelated, so every uncertainty figure in the draft is a floor rather than an estimate. The operational analysis uses an AR(1) model with $\rho = 0.4$ and derives the exact daily-mean variance.	2–3 sentences + caveat
5	Bulk–wall compensation	k_b is fixed at -0.5 day $^{-1}$ and excluded from the ensemble, yet it is the parameter that trades off against k_w . Any error in it is absorbed into the wall coefficients and into the risk map.	~ 0.5 page: repeat GLUE at $k_b \pm 20\%$
6	Held-out validation	The draft's key claim is extrapolation of risk to 86 unmonitored junctions. Leave-one-monitor-out validation is the standard evidence for that claim and is cheap with six monitors.	~ 0.5 page + 1 table
7	Toolchain verification	A known-answer test (single pipe with an analytic first-order solution) establishes that the coefficients are realised as intended. One paragraph buys considerable credibility.	1 paragraph
8	± 0.1 mg/L convention	The draft treats ± 0.1 as one standard deviation. Read as a 95% band it implies $\sigma \approx 0.05$ and halves every interval. State the convention explicitly.	1 sentence

5. Priority 3 — framing of the risk product

The draft's final product is described as a low-chlorine risk map, and the Discussion states that it identifies where intervention “would be considered first in a real-network application”. Two corrections of framing are needed, both drawn from the operational analysis.

First, name the outcome correctly. It is a sensor-conditioned probabilistic forecast from a network model, not a spatial measurement and not geostatistical interpolation. Its credibility is conditional on the

hydraulic model, the boundary conditions, the sensor QA state, the kinetic structure and the prior ranges. It is not a statement about whether water is safe: chlorine residual is one operational indicator among the preventive barriers of a Water Safety Plan. A single sentence stating the conditioning clause, and one stating what the map is not, would markedly improve the Discussion.

Second, the time-averaged probability conflates two different operational situations. A node that sits marginally below the threshold at all times and a node that falls far below it for two hours a day can share the same P value but demand different responses. Report either the exceedance duration or the depth of the excursion alongside P, at least for the highest-ranked nodes.

If space permits, a short subsection on how the calibrated ensemble would be operated as new data arrive would connect the dissertation to the project's operational strand: a fast surveillance loop holding the model fixed and converting new measurements into innovations for sensor QA and event detection, and a slow loop in which recalibration occurs only behind sensor-QA and hydraulic verification gates, with an explicit statement of which parameters the data can and cannot refresh. One page and one schematic, no more — and only after §3 is resolved.

6. Length, structure and presentation

6.1 Length

The draft renders at 59 pages against a 30-page limit: A4, 12 pt, 1.5 line spacing, 3.2 cm side margins, giving roughly 200 words per page. Reaching 30 pages requires both reformatting and cutting; neither alone is sufficient.

Action	Approximate saving
Line spacing 1.5 → 1.15	12–14 pages
Side margins 3.2 → 2.5 cm	4–5 pages
Reduce Background from ~2,900 to ~1,500 words (compress the chlorine-chemistry primer; retain the identifiability and GLUE-critique material)	3–4 pages
Convert numeric prose to four tables: grid-search recovery, GLUE summary, noise sensitivity, risk ranking	1–2 pages, and clearer
Combine figure pairs into two-panel figures (zoned/interleaved maps; zoned/interleaved GLUE distributions; the two predictive bands), and crop white space	2–3 pages
Move the unit-conversion arithmetic and the 60 s/300 s timestep check to an appendix	~1 page

Confirm whether the 30-page limit includes references and appendices; the plan above assumes it applies to the main body.

6.2 Structure and presentation

- No Word heading styles are used anywhere in the document, so there is no navigation pane, no automatic table of contents and no reliable numbering. Apply Heading 1/2/3 styles throughout and insert a generated contents page.
- Section numbering is inconsistent: some headings are numbered (3.5.1, 3.6.2), most are not, yet the text cross-references “Section 3.4”, “Results Section 4.1” and similar. Number every section and make the cross-references match.

- No figure is numbered. Every caption reads “Figure.”, so no figure can be cross-referenced from the text. Number all 15, add a list of figures, and reference each one at the point of discussion.
- The document contains no tables at all. Several passages of dense numeric prose in the Results would be substantially clearer, and shorter, as tables.
- Placeholders remain: the Acknowledgements section reads “a....” and the Appendices heading has no content beneath it.
- Front matter is missing: title page with author, supervisor, degree, date and word count; declaration; contents; list of figures; nomenclature table for the symbols used in §3.
- Minor consistency: “behavior” appears in one Results heading against UK spelling elsewhere; “this thesis” and “this study” are used interchangeably; there is a doubled “in in this thesis” in §3.2.

6.3 References

The reference list is appropriate in scope, but a dissertation whose main method is GLUE should engage with the statistical critique of GLUE rather than only its originating paper. Add at least Stedinger et al. (2008) and Mantovan and Todini (2006), and respond to them briefly in the Background. The draft's own finding that the behavioural ensemble is close to the prior is a concrete instance of what those papers warn about, which makes the engagement easy and pointed. Beven (2006) is already cited. Consider adding Powell et al. (2000) for bulk decay.
